

Prince Acquah Rockson

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Profile

Researcher in Health Informatics and Machine Learning, specializing in clinical risk prediction and disease modeling. Experienced in developing interpretable intelligent systems for non-communicable diseases such as chronic disease staging (CKD). Committed to advancing scalable, privacy-preserving AI to bridge the gap between clinical data and global health interventions.

Education

Kwame Nkrumah University of Science and Technology

HEALTH INFORMATICS, Master of Science

Kumasi, Ghana

Jan. 2024 - Nov. 2025

University of Cape Coast

MOLECULAR BIOLOGY AND BIOTECHNOLOGY, Bachelor of Science

Cape Coast, Ghana

Sept. 2014 - May 2018

Research Experience

- **Topic:** Development and Validation of an Explainable Framework for 6-Class Chronic Kidney Disease Staging: A Computational Diagnostic Study
 - **Status:** Manuscript submitted to BMC Medical Informatics and Decision Making (Awaiting Preprint)
 - **Key Contribution:** Engineered a multi-stage diagnostic framework using Gradient Boosting Machine, LightGBM, XGBoost, and others. Implemented 100-iteration bootstrapping for stability and SHAP for clinical interpretability.
 - **Role:** Principal Researcher / Lead ML Architect.
 - **Advisor:** Dr. Eric Osei Opoku
- **Topic:** A Performance and Interpretability Study of Machine Learning Models for SYN Flood Detection (2025)
 - **Focus:** Cybersecurity & Trustworthy AI.
 - **Key Contribution:** Comparative analysis of ML models with a focus on Model Interpretability. Designed feature selection pipelines to identify critical network attack signatures.
 - **Advisor:** Kwame Owusu Kwarteng
- **Topic:** Improving Investment Optimization Under Inflation and Sovereign Debt Constraints in the Ghanaian Market Using Machine Learning (2025)
 - **Focus:** Time-series Forecasting & Economic Modeling.
 - **Key Contribution:** Developed predictive models for financial optimization in volatile markets (Ghana). Handled complex, high-dimensional economic indicators.
 - **Advisor:** Prof. Benjamin Ghansah.
- **Topic:** Application Of Machine Learning Techniques for Detecting Fraudulent Transactions in Halges Fintech (2025)
 - **Focus:** Anomaly Detection & Financial Security.
 - **Key Contribution:** Developed and deployed predictive models to identify fraudulent patterns in high-frequency transactional data at **Halges Fintech**.
 - **Advisor:** Warlali Ametepe.
- **Topic:** Detection Of Genetically Modified Products Using PCR(2018)
 - **Focus:** Biotechnology / Molecular Biology
 - **Key Contribution:** Performed laboratory-scale genomic analysis; established the "wet-lab" foundation that informs my current understanding of molecular health data.
 - **Advisor/Supervisor:** Dr Emmanuel Plas Otwe

Training and Certifications

Centre For Data Science and Analytics/ Ashoka University

Virtual

Data Analytics For Climate and Health Azubi Africa	Sep. 2025 <i>Virtual</i>
Data Analytics Professional Coursera	Aug. 2024 <i>Virtual</i>
Google Data Analytics Amalitech	May. 2024 <i>Virtual</i>
Software Development Programme Coursera	Sep. 2023 <i>Virtual</i>
Google Project Management Coursera	Aug. 2023 <i>Virtual</i>
Google IT Support Centre For Business Analysis	Jun. 2023 <i>Accra, Ghana</i>
Certified Business Data Analyst Google Africa Developer Training/ Andela	Aug. 2024 <i>Virtual</i>
Android Development	May. 2024

Professional Experience

Halges Financial Technologies (Korba) *Accra, Ghana*
 Data Analyst(BI and Audit Functions)
 June 2024 - Present

- Collect, clean, and analyze large transactional datasets to identify trends, anomalies, and performance gaps.
- Prepare and present monthly executive performance reports to support strategic decision-making.
- Partner with product and finance teams to generate actionable insights that drive operational efficiency.
- Lead daily reconciliation of wallet, bank, and merchant transactions to ensure financial accuracy and system integrity.
- Investigate and resolve transaction discrepancies, improving settlement accuracy with external partners.
- Design and maintain automated reporting dashboards using Excel and SQL, reducing manual reporting time.
- Detect data inconsistencies and strengthen database integrity through validation checks and audit controls.

Docupro Ltd *Accra, Ghana*
 Lead Quality Assurance Officer,
 May 2020 - June 2024

- Oversaw weekly data quality assurance across multiple client databases and systems.
- Developed monthly validation scripts to detect and correct inconsistent or missing data.
- Monitored data pipelines daily to ensure accuracy, completeness, and timeliness.
- Collaborated with engineers to improve data integration and governance workflows.
- Prepared periodic data quality reports and recommended process enhancements.

S.K. Consult *Cape Coast, Ghana*
 Associate Consultant
 Mar 2020 - Dec. 2023

- Collaborated with at least 100 clients to define their research topics, objectives, and scope.
- Collected quantitative and qualitative data through surveys and interviews for at least 100 clients.
- Developing and selecting appropriate research plans, research methods, and data analysis tools.
- Analysed and interpreted both primary and secondary quantitative/qualitative data using statistical tools like SPSS, R, Stata, PLS-SEM, and analytical techniques tools like t-tests, MANOVA, and ANCOVA.
- Generated reports and presented findings with appropriate visual representations

Parockson Digital Solutions *Accra, Ghana*
 Head of Projects
 March 2019 - Present

- Leading end-to-end planning and execution of software and AI research projects.

- Supervising cross-functional teams to deliver client solutions on schedule.
- Managing project budgets, timelines, and stakeholder communications.
- Oversaw the design and rollout of digital solutions for clients.
- Ensured quality assurance and documentation for all completed projects.

Biotechnology and Nuclear Agriculture Research Institute

Accra, Ghana

Research Assistant and Head of BSF lab (NSS and Volunteer)

Sep. 2018 - May. 2020

- Led Black Soldier Fly (BSF) laboratory operations and experimental setups.
- Collected, analyzed, and interpreted data from BSF growth and waste conversion studies.
- Supervised other research assistants and ensured adherence to lab safety protocols.
- Prepared technical reports and contributed to research publications.
- Supported project design and data analysis for agricultural biotechnology research.

Daasgift Quality Foundation

Takoradi, Ghana

Monitoring & Evaluation (M&E) Intern

June 2017 - Aug. 2017

- Collected and analyzed project data to track performance indicators.
- Assisted in preparing weekly M&E reports and documentation for donor submissions.
- Supported field data collection and beneficiary feedback sessions.
- Helped design survey tools and evaluation templates for ongoing projects.
- Contributed to data entry, cleaning, and visualization for impact assessments.

Peer Review Profiling

- ACM RESPECT 2025 Conference (Association for Computing Machinery) | Jan 2025 - Present
 - Evaluating research submissions focused on equity and computer science education research.
- Healthcare Informatics Journal | June 2024 - Present
 - Reviewing manuscripts on the application of IT, data science, and ML in healthcare delivery.
- American Society for Engineering Education (ASEE) | June 2024 - Present
 - Peer-reviewing conference papers regarding engineering methodology and technical curriculum.

Workshops, Webinars & Conferences

Faculty of Physical and Computational Sciences PosterFest | Aug 2025 | Kumasi, Ghana

- Presenter: "Detection and Staging of Chronic Kidney Disease: A Step Toward Preventive Nephrology."
- Highlighted the integration of Machine Learning for subclinical diagnosis in resource-limited settings.

Centre for Business Analysis (Virtual) | Aug 2025

- Invited Speaker: "The Future of Data Analytics: How Certifications Keep Us Relevant."
- Discussed the evolving landscape of AI and the necessity of continuous skill acquisition in data science.

ASPIRE 2025 Program (Virtual) | Aug 2025

- Keynote Speaker: "Telling the Data Story."
- Trained emerging professionals on data visualization and communicating complex insights to non-technical stakeholders.

Bayobab Africa (Virtual) | Aug - Nov 2025

- Technical Speaker: "ITIL Frameworks and Cloud Computing Service Life Cycles."
- Delivered a modular series on service improvement, stakeholder management, and cloud infrastructure roles.

Leadership Roles & Experiences

Change Advisory Board Member

Accra, Ghana

Halges Financial Technologies Ltd

Mar., 2026.- Present

- Risk Assessment: Evaluating how a proposed update might affect existing workflows or data integrity
- Stakeholder Alignment: Ensuring that changes to communication channels or branding assets maintain the established visual identity and professional reputation.

- **Schedule Coordination:** Reviewing the "Change Calendar" to ensure updates do not conflict with major events.
- **Post-Implementation Review (PIR):** Analyzing whether a change achieved its desired outcome and documenting lessons learned to improve future deployment cycles.

User Acceptance Testing (UAT) Team Member

Accra, Ghana

Halges Financial Technologies Ltd

, July., 2024.- Present

- **Test Case Development:** Creating detailed scenarios based on user requirements (e.g., "Can a researcher successfully upload a dataset for ML staging?").
- **User Feedback Synthesis:** Managing pilot groups of alumni to gather qualitative data on UI/UX and translating that into actionable bug reports for developers.
- **Defect Management:** Using tools like GitHub issues to track functional gaps, ensuring the final product reflects the intended branding and professional standards.
- **Final Sign-off:** Providing the formal "Go/No-Go" recommendation before a platform or tool is launched to the wider network.

Team Lead

Accra, Ghana

Jospong ICT Data Science Team for Shared Services

Jan., 2024.- Present

- Led a cross-functional data science team delivering analytics and automation solutions for shared services units
- Oversaw end-to-end project execution, from problem definition and data collection to model development, validation, and deployment
- Guided junior analysts through technical reviews, mentoring, and skill development
- Trained and mentored at least 15 data analysts and scientists for shared services.

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Skills and Languages

Computational Research & Data Science

- **Languages:** Python (Advanced: NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn), R (Statistical Modeling), SQL (PostgreSQL, MySQL).
- **Statistical Analysis:** SPSS, Stata, PLS-SEM, Excel.
- **Specialized AI:** Machine Learning Model Validation (Bootstrapping, Cross-Validation), Explainable AI (SHAP/LIME).

Software Engineering & Deployment

- **Web & App:** React JS/TS, Node.js, Streamlit (for ML Dashboarding).
- **DevOps & Infrastructure:** Git/GitHub, Netlify, Render, Railway, Vercel.
- **IT Architecture:** ITIL Frameworks, Cloud Computing Service Lifecycle.

Data Visualization & Business Intelligence

- Power BI, Advanced Excel (VBA/Macros), Tableau.

Languages

- **English:** Native / Professional Fluency (Full Research Proficiency).
- **Local Languages:** Asante Twi, Fante, Ga.